GOOGLE TRENDS AND FINANCIAL MARKETS

Exploratory Data Analysis of “FII Buying” Search Interest and NIFTY 50 Returns

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Abstract

This report is trying to find whether Google Trends search interest for keywords (Eg used in the notes: “FII Buying”) can provide useful information to trade the Index NIFTY 50. The notebook fetches NIFTY 50 closing prices using yfinance and search-interest observations using pytrends. Calculates daily percentage returns cleans the data and then explore descriptive statistics, correlation, scatter plots and regression lines to draw a conclusion on the usefulness of Keywords in forecasting Nifty 50 trend.

1. Introduction

Financial researchers today are looking beyond traditional data sources like company financials and stock prices to generate alpha from the markets. This is where alternative data comes into play non-traditional information such as satellite images, credit card transactions, social media posts, and online search trends can provide useful information into the markets. For Eg: if a person is getting interested in a topic he might search it up online to get to know more about it. This makes search activity of people a useful information to track. A systematic review of 56 studies found that Google Search Volume Index research has examined returns, volatility, trading volume, and forecasting, while also noting that findings differ across keywords, regions, and data frequencies (Ayala et al., 2024).

This project studies how search keywords (“FII Buying in this case”) compared against NIFTY 50. Google Trends reflect the relative popularity on a scale of 0-100 as opposed to raw number of seacrhes. As a result, higher search results indicate curiosity and the actual trading volumes by Foreign Investors.

2. Research Question and Objectives

Research question: Does Google Trends search interest or its change over time, show a useful relationship with daily NIFTY 50 returns?

- Fetch NIFTY 50 price data and Google Trends interest data for India.
- Clean and align the datasets by date.
- Calculate daily percentage returns and changes in search interest.

3. Data and Methodology

3.1 Data sources and sample

The notebook uses the keyword “FII Buying”. Google Trends geography as India (IN), ticker as ^NSEI, and sample start date as 1st January 2022. The price data is fetched through yfinance and trends data through pytrends. Since the data is being fetched from 2 different sources the data is first cleaned and then aligned to perform further descriptive statistics, correlation and regression analysis.

3.2 Variables

- close: adjusted NIFTY 50 closing price returned by yfinance.
- return_pct: percentage change in close from the previous day close.
- interest: normalized Google Trends index for “FII Buying.”
- interest_change: first difference in the interest index.

3.3 Alignment and analysis

The notebook uses backward as of merge to align with market date with the trends on respective date. It then calculates daily returns in precentage and the first difference in aligned interest. The analysis includes summary statistics, a correlation matrix, a time-series plot of interest and close, a scatter plot of interest versus returns, and regression plots comparing returns with interest level and interest change.

4. Results and Interpretation

The interpretation of the notebook suggests that regression line for interest level versus daily NIFTY 50 returns is nearly flat, with a slight upward slope. The observations all fall between the range of 20-40. Which suggests the relationship is too weak to conclude any meaningful conclusion. The correlation table analysis also found a very weak positive relationship between interest level and NIFTY 50 returns. Pearson correlation was 0.0394 ($p = 0.1762$), while Spearman correlation was 0.0431 ($p = 0.1385$), based on 1,182 observations.

On the other hand, change in search interest, the regression line is more noticeably upward-sloping. But the scatter plots suggest that most of the observations are focused around the middle. Pearson correlation was 0.0568 ($p = 0.0508$) and Spearman correlation was 0.0193 ($p = 0.5069$), also based on

1,182 observations. Suggesting that even though the correlation is positive it is too weak. Thus, concluding that the relationship is not sufficient to establish that “FII Buying” interest can forecast future NIFTY 50 returns.

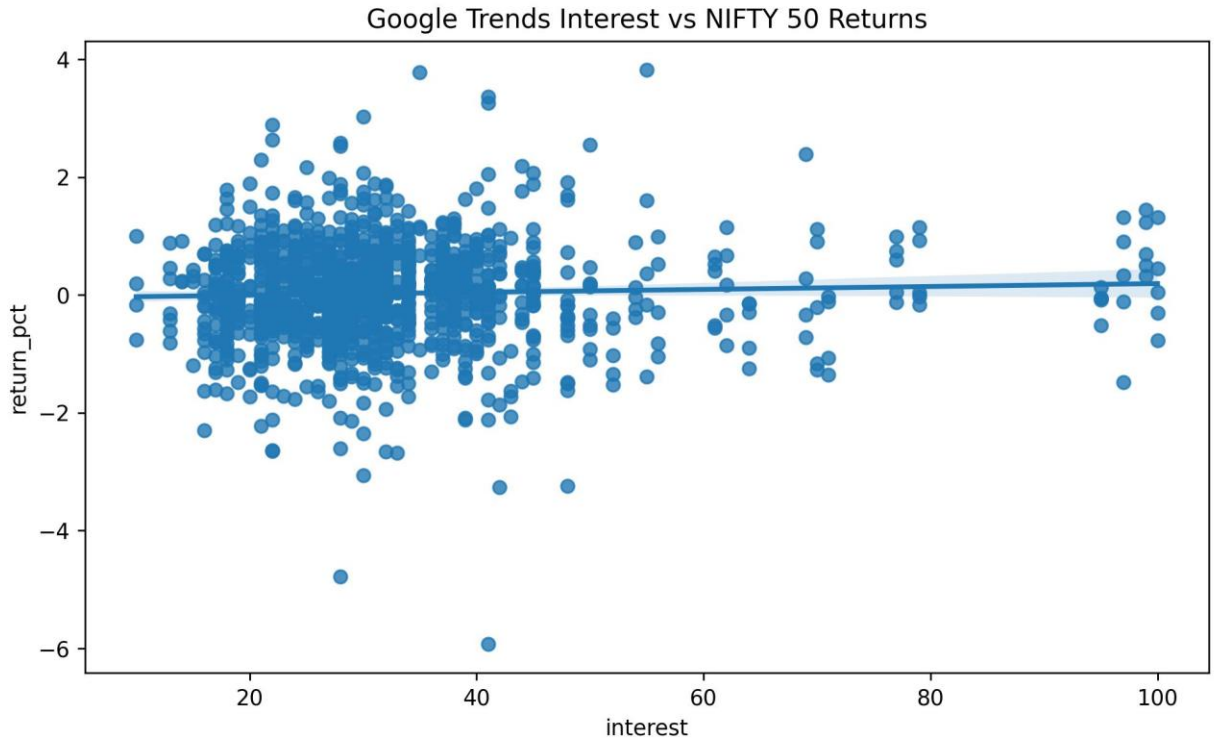
5. Limitations

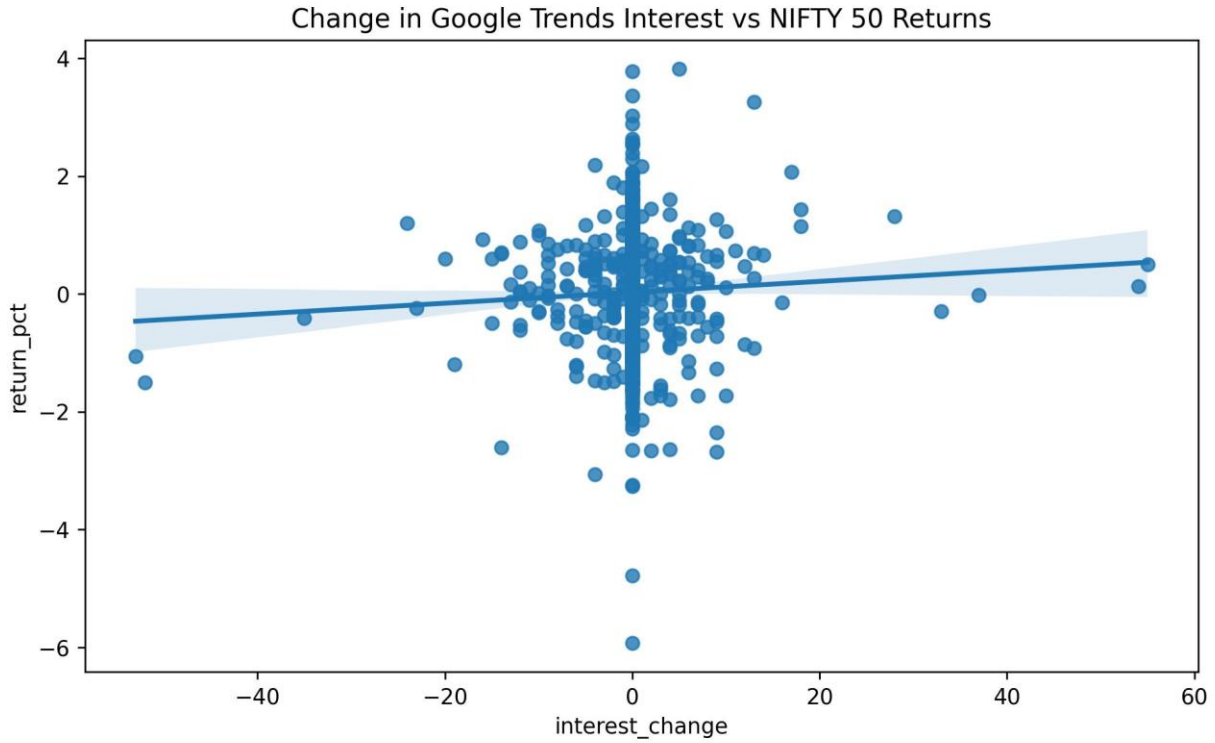
- Correlation and regression slopes do not establish causation.
- Searches for “FII Buying” may reflect news, curiosity, education, or uncertainty rather than actual institutional flows.
- Google Trends is normalized and sampled, not an absolute search count (Google, n.d.).
- The case study uses one query; its results cannot be generalized to all search keywords.

6. Conclusion

The project explores whether search interest for search queries relate to daily NIFTY 50 returns. The case study uses “FII Buying” to investigate NIFTY 50 returns. The 80/20 test split used 944 observations for training and 237 observations for testing, covering 31 October 2025 to 16 October 2026. The linear regression model achieved a slightly lower MAE than the baseline but its RMSE and R-squared were worse. The differences were small and do not indicate a useful predictive advantage. The overall conclusion is even though the correlation is positive it is not strong enough to have any meaningful impact on the results. Note that this result is based on the keyword “FII Buying” and the results may vary for different keywords used.

7. Outputs


Google Trends Interest vs NIFTY 50 Returns
return_pct
interest
4
2
0
−2
−4
−6
20
40
60
80
100


Change in Google Trends Interest vs NIFTY 50 Returns
return_pct
interest_change
4
2
0
−2
−4
−6
−40
−20
0
20
40
60

Correlation Results

Variables	Sample size	Pearson correlation	Pearson p-value	Spearman correlation	Spearman p-value
interest vs return_pct	1182	0.0393	0.177	0.0451	0.1214
interest_change vs return_p	1182	0.0523	0.0723	0.009	0.7573

Train/Test Results

Model	Test observations	MAE	RMSE	R-squared
Linear regression	237	0.653	0.8785	-0.0106
Mean-return baseline	237	0.6532	0.8781	-0.0098

8. References

- Ayala, M. J., Gonzálvez-Gallego, N., & Arteaga-Sánchez, R. (2024). Google search volume index and investor attention in stock market: A systematic review. *Financial Innovation, 10*, Article 70. https://doi.org/10.1186/s40854-023-00606-y
- Google. (n.d.). FAQ about Google Trends data. Google Trends Help. Retrieved October 17, 2026, from https://support.google.com/trends/answer/4365533
- GeneralMills. (n.d.). pytrends: Pseudo API for Google Trends. GitHub. https://github.com/GeneralMills/pytrends
- Yahoo Finance. (n.d.). NIFTY 50 (^NSEI) historical data. Retrieved October 17, 2026, from https://finance.yahoo.com/quote/%5ENSEI/history/